Ouissal Chaib Amensou

Computer Science Engineering Student

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PROFILE

As a Computer Science Engineering and Business Management dual-degree student, I have completed multiple programming projects, gaining solid experience in various languages and technologies. Motivated by continuous learning, I have a strong and growing interest in Artificial Intelligence, Cryptography, and Cybersecurity. I am seeking a 4-month research internship where I can apply my current expertise, contribute to meaningful academic projects in AI or Security, and further strengthen my technical and cross-cultural abilities in an enriching environment.

TECHNICAL SKILLS

Security & Networking: Cryptography, TCP/IP, Wireshark, GDB, Valgrind, Memory Safety basics

Programming Languages: C, C++, Python, Java, Assembly, JavaScript, HTML/CSS

Frameworks & Tools: Git, Linux, Bash, SQL/PostgreSQL, Arduino, MATLAB

Methodologies & Concepts: Artificial Intelligence, Graph Theory, Algorithm Design, Agile, Project Management

EDUCATION

Master of Engineering in Computer Science *2025 – 2028*

ENSICAEN – National Graduate School of Engineering, Caen, France

Coursework: AI, Cryptography, Graph Algorithms, Software Engineering, Networks, Embedded Systems, Web Dev, Programming Languages, Advanced Mathematics, Economics & Management.

Business Management Diploma (dual degree) *2025 – 2028*

EM Normandie Business School, France

Preparatory Classes for Engineering Schools (CPGE), Physics & Engineering Sciences *2022 – 2025*

Lycée Chaptal, Paris, France

Coursework: Mathematics (Linear Algebra & Analysis), Physics, Engineering Sciences, Computer Science

French Baccalaureate (High School Diploma) — Highest Honors *2019 – 2022*

French High School Diploma — Mathematics, Physics-Chemistry, Economics & Social Sciences

ACADEMIC PROJECTS

GrandPrix — Autonomous Racing Pilot (C)

- Engineered an autonomous racing AI in C using an A* pathfinding algorithm with a custom heuristic incorporating inertia, velocity projection, and fuel constraints.
- Implemented a binary min-heap priority queue for efficient state retrieval, alongside real-time trajectory validation to ensure safe navigation and collision avoidance within a Qt/C++ environment.

Color Histogram & Image Quantization (C)

- Optimized memory usage by replacing massive 3D arrays with a 256x256 2D array of linked lists to compute RGB color frequencies for PPM images.
- Developed a popularity-based color quantization algorithm mapping pixels to K dominant colors, alongside a custom command-line visualization tool.

Directory Database Management System (C)

- Developed a C application to manage a directory database, implementing core CRUD operations.
- Designed a structured record-storage system with strict input validation to prevent buffer overflows and ensure data integrity.

Web Project – HTML/CSS/JavaScript Animation

- Created a dynamic, interactive web animation using vanilla HTML, CSS, and JavaScript.

PROFESSIONAL EXPERIENCE

Private Math Tutor *2024 – 2026*

Caen, France

- Tutored 8 high school students in mathematics and physics delivering personalized instruction on coursework and exam preparation that improved grades by 2–5 points.

Elected Student Representative *2025 – 2026*

ENSICAEN, Caen, France

- Elected to the School's Board of Directors to vote on budget allocations and monitor school-wide academic projects.
- Served on the Disciplinary Committee, analyzing student cases and facilitating communication between students, faculty, and administration.

Treasurer, Sports Committee (Bureau des Sports) *2025 – 2026*

ENSICAEN, Caen, France

- Managed and monitored the committee's overall budget, negotiating vendor contracts to control operating costs.
- Led and coordinated a 20-member team, organizing recurring meetings and large-scale campus athletic events.

ADDITIONAL INFORMATION

Languages: French (Native), English (B2+), Spanish (B2+), Arabic (B1), Japanese (Beginner – eager to learn)

Certifications: PCS1 – Basic First Aid & Civil Safety Certificate

REFERENCES

Prof. Christophe Rosenberger: AI & Cybersecurity, Director of GREYC Lab | christophe.rosenberger@ensicaen.fr

Prof. Hugo Descoubes: Electrical Engineering, Head of Embedded Systems | hugo.descoubes@ensicaen.fr